Introduction to Machine Learning and Artificial Intelligence

Model Evaluation and Model Validation

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Overview

- **Model Evaluation**
- **Model Validation**

Model Evaluation

Introduction

- Model evaluation can be performed in **supervised learning** as the actual values are available
- there is a fundamental difference between the methods used for evaluating a regressor and a classifier:
  - in regression, we deal with continuous values and the focus is on the **error between the actual and predicted values**
  - in classification, the focus is on the **number of data that are classified correctly**

- In classification, we deal with two types of models:
  - in some classifiers (e.g., k-NN, SVM), **the output is simply the class label**
  - in others (e.g., logistic regression, random forest), **the output is the probability of a data point belonging to a particular class** where through the use of a cut off value we are able to convert these probabilities into class labels

Source: www.techplayon.com/

Confusion Matrix

Suppose a data set with class labels 0 (‘negative’ class) and 1 (‘positive’ class), and a classifier that can produce correct and incorrect predictions:

- **True positive:** the number of positive observations that were correctly predicted by the model
- **True negative:** the number of negative observations that were correctly predicted by the model
- **False positive:** the number of negative observations that were incorrectly predicted as positive by the model
- **False negative:** the number of positive observations that were

		Predicted Class		
		Positive	**Negative**	
Actual Class	**Positive**	True Positive (TP)	False Negative (FN) Type II Error	**Sensitivity** $\frac{TP}{(TP+FN)}$
	Negative	False Positive (FP) Type I Error	True Negative (TN)	**Specificity** $\frac{TN}{(TN+FP)}$
		Precision $\frac{TP}{(TP+FP)}$	**Negative Predictive Value** $\frac{TN}{(TN+FN)}$	**Accuracy** $\frac{TP+TN}{(TP+TN+FP+FN)}$

Our objective is to minimize the FP and FN and maximize the TP and TN, that is, to maximize the diagonal values

Source: www. encord.com

Other Evaluation Measures

Accuracy and classification error can be very misleading
For instance, suppose a data set with 100 instances where 99 belong to class 'A' and only 1 belongs to class 'B'. A model that predicts all instances to class 'A' will have an accuracy of 99%, but the accuracy here provides us with little information on how actually the model is performing

- **False positive rate (false alarm):** the proportion of negative observations that were wrongly (as positive) predicted by the mode

$$FPrate = \frac{FP}{FP + TN}$$

- **False negative rate (miss rate):** the proportion of the positives that were classified as negative by the model

$$FNrate = \frac{FN}{FN + TP}$$

- **Precision (positive predicted value):** the proportion of the positives compared to what the model predicted as positive. Unlike TPrate, Precision provides the percentage of TP over all instances predicted as positive

$$Precision = \frac{TP}{TP + FP}$$

Source: www.techplayon.com/

Summary Evaluation Metrices

Metric	What it means	Why it is important	What's a good value	Simple Example (Heart Disease Prediction)
Accuracy	How often the model's predictions are correct overall. Formula: (TP + TN) / (TP + TN + FP + FN)	Shows general performance, good when both classes (e.g., "disease" and "no disease") are about equal.	**Higher is better** (close to 1 or 100%)	The model correctly predicts 90 out of 100 patients → **Accuracy = 0.90 (90%)**
AUC (Area Under the ROC Curve)	Tells how well the model separates the two classes (positive vs. negative).	Measures how well the model can **tell one class from the other**, even if data is unbalanced.	**Higher is better** 0.5 = random guessing, 1.0 = perfect	AUC = 0.95 The model can tell healthy vs. sick patients correctly 95% of the time.
Sensitivity (Recall / True Positive Rate)	How many of the actual positive cases were correctly found. Formula: TP / (TP + FN)	Important when **missing positive cases is dangerous** (like missing a disease).	**Higher is better** (but if too high, might increase false alarms)	Out of 100 sick patients, the model finds 85 correctly **Sensitivity = 0.85**
Specificity (True Negative Rate)	How many of the actual negative cases were correctly found. Formula: TN / (TN + FP)	Important when **false alarms are bad**, like giving treatment to a healthy person.	**Higher is better** (ideally both Sensitivity and Specificity are high)	Out of 100 healthy patients, 90 are correctly marked healthy **Specificity = 0.90**

How to Understand Together

Situation	What it means	What to do
High Accuracy but Low AUC	Model seems good but can’t really tell classes apart.	Check if your data is **imbalanced** or if model is **overfitting**.
High Sensitivity, Low Specificity	Catches almost all real positives but also many false alarms.	OK for **medical screening**, where missing a disease is worse than a false alarm.
High Specificity, Low Sensitivity	Avoids false alarms but misses real positive cases.	OK for **fraud or security**, where false accusations are costly.
High Accuracy + High AUC	Model performs well and clearly separates both classes.	Ideal: balanced and reliable model.

How to Understand Together

Metric	Decision Tree	Random Forest	What It Means
Accuracy	0.82	0.88	Random Forest makes more correct predictions overall.
AUC	0.85	0.93	Random Forest separates healthy vs. diseased better.
Sensitivity	0.78	0.86	Random Forest detects more sick patients.
Specificity	0.84	0.89	Random Forest identifies healthy patients more accurately.

Regression Models

In regression, evaluation methods depend on the ways of calculating the residual (difference between the actual and predicted values)

–some methods (e.g., sum squared error, mean squared error and root mean squared error) just calculate this difference

–others (e.g., adjusted coefficient of determination) also take into account the problem of overfitting

In regression, unlike classification (where we can count the number of observations correctly classified), our predictions are either bigger or smaller than the actual value (rarely it is same as the actual value)

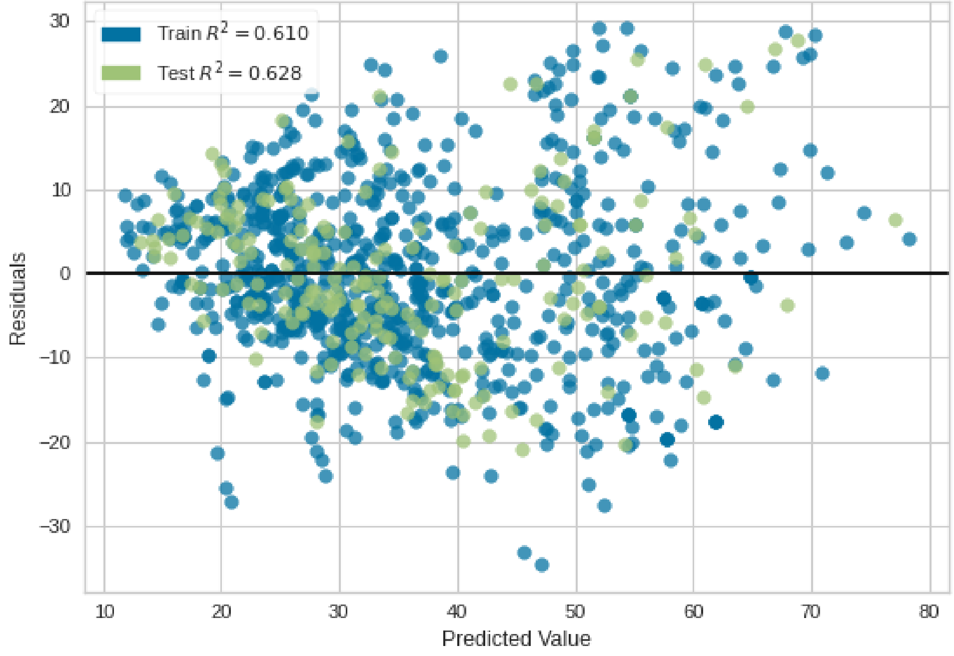
Thus, we are not concerned with how many times we were wrong but rather what matters is the size of residuals

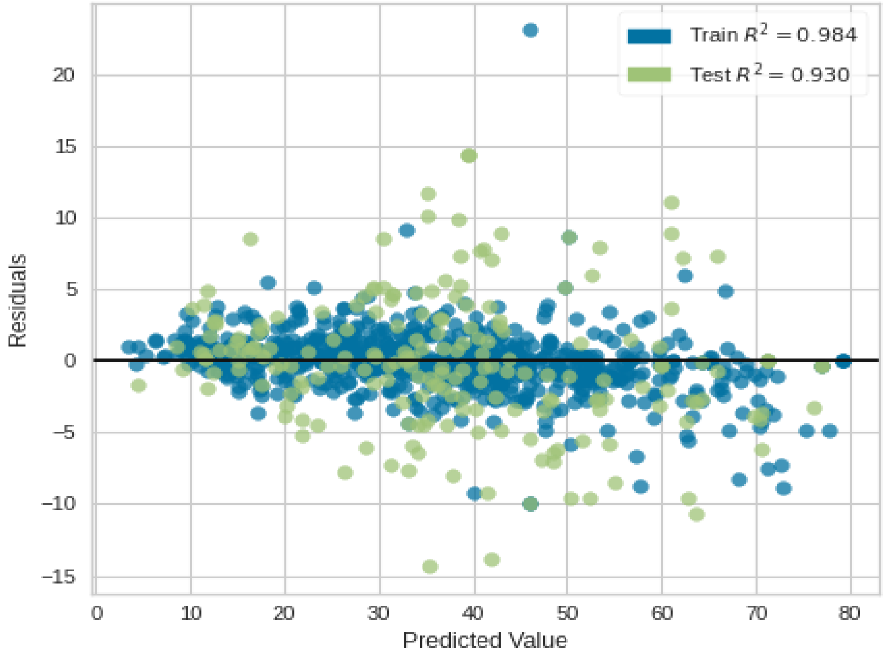
To validate a regression model, you must use residual plots to visually confirm the validity of the model

Regression Models

Residual plots: the residual values on the y-axis and the predicted values on the x-axis

Residual = Observed Value - Predicted Value

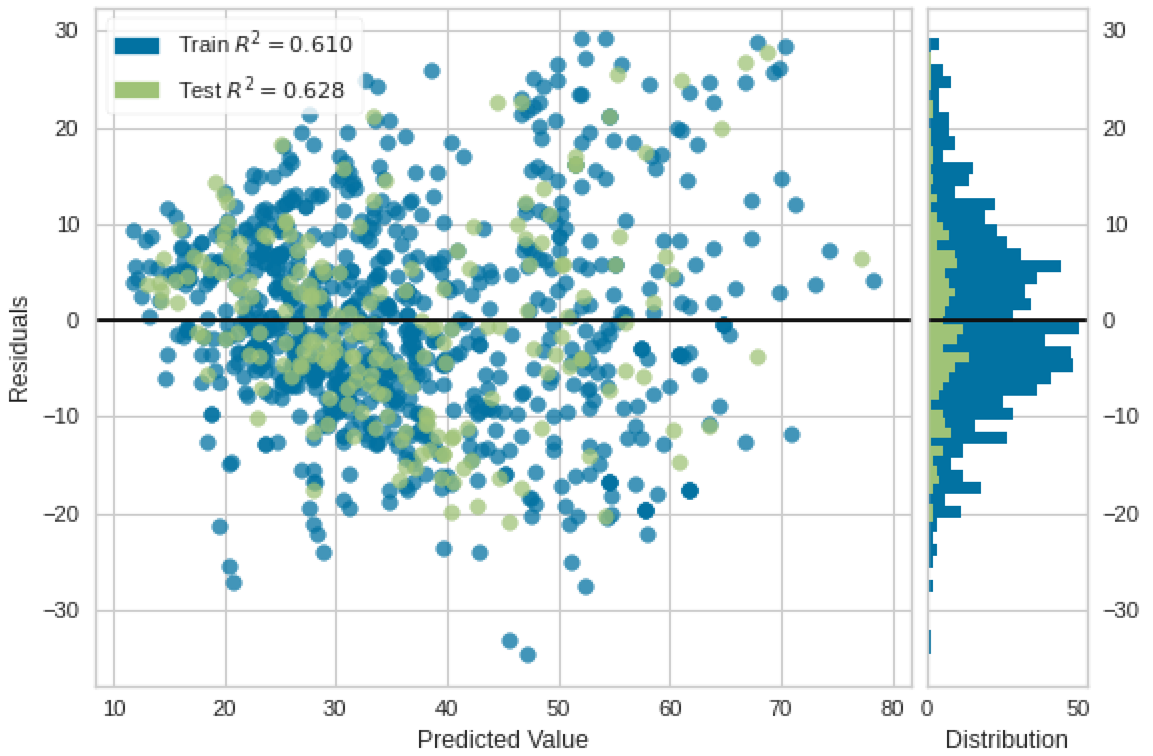


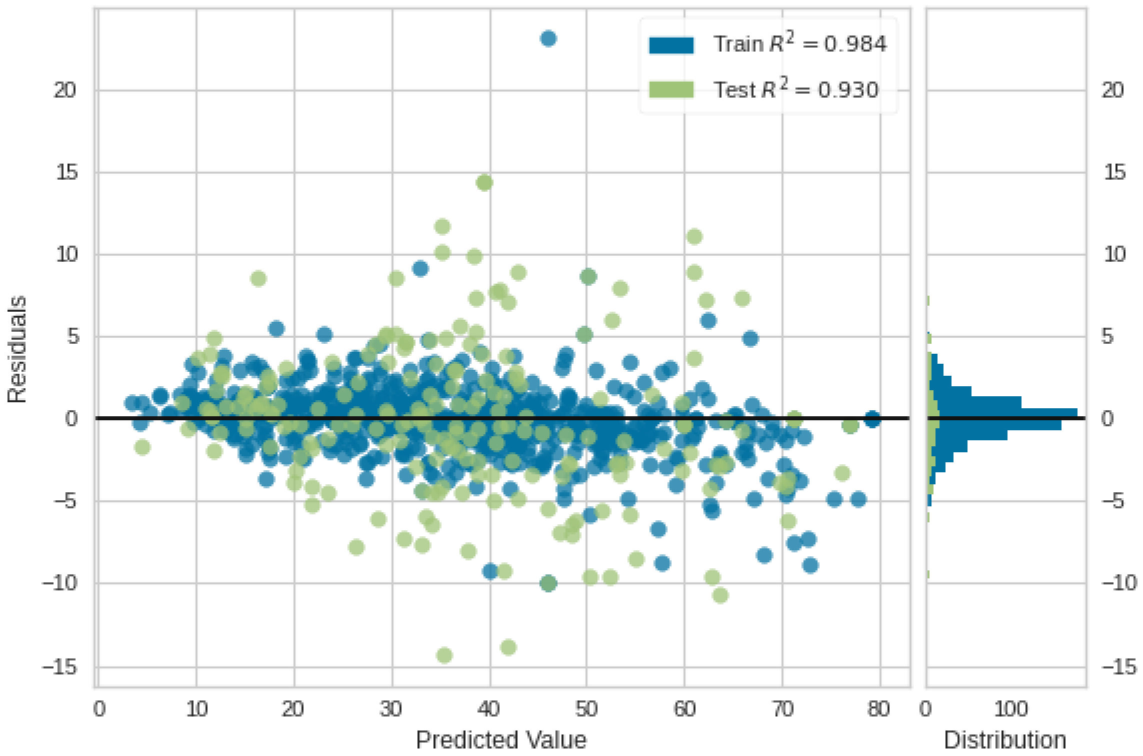


Source: www. encord.com

Regression Models

How to know whether a residual plot is good based on thecharacteristics just presented?





If we project all the residuals onto the y-axis, we get a normally distributed curve.
This satisfies the assumption that **residuals are independent and normally distributed.**

Metric Selection

Central Tendency Measures

Mean of Prediction: Arithmetic average of predicted values

$$\mu = \frac{1}{n} \times \sum_{i=1}^{n} \hat{y}_i$$

Population Mean Formula

Median of Prediction: Middle value when predictions are sorted

μ = Population mean
n = Number of observations
$\hat{y}_i$ = Predicted value for observation i

- Mean helps understand prediction bias between models
- Median is robust to outliers
- Use median when data contains extreme values

Standard Deviation

Measures dispersion of predicted values around the mean

$$\sigma = \sqrt{\left[\frac{1}{n} \times \sum_{i=1}^{n} (\hat{y}_i - \mu)^2\right]}$$

Population Standard Deviation

$$s = \sqrt{\left[\frac{1}{n-1} \times \sum_{i=1}^{n} (\hat{y}_i - \bar{x})^2\right]}$$

Sample Standard Deviation

Low SD
Values clustered near mean

High SD
Values spread over wider range

SD quantifies prediction consistency - lower values indicate more stable predictions

Range of Prediction

Simple measure of data dispersion using extreme values

$$R = \hat{y}_{max} - \hat{y}_{min}$$

Range Formula

$\hat{y}_{max}$ = Maximum predicted value
$\hat{y}_{min}$ = Minimum predicted value
R = Range

- Provides quick insight into prediction variability
- Sensitive to outliers
- Useful for initial data exploration

Metric Selection

Coefficient of Variation

Relative standard deviation for comparing different datasets

$$CV = \sigma/_{\mu} \times 100\%$$

Population Coefficient of Variation

$$CV = s/_{\bar{x}} \times 100\%$$

Sample Coefficient of Variation

σ = Population standard deviation
s = Sample standard deviation
μ = Population mean
x̄ = Sample mean

- Standardizes variance by the mean
- Enables comparison between datasets with different scales
- Useful when means differ significantly
- Expressed as percentage for intuitive interpretation

Key Insight: SD measures single dataset variability, while CV enables cross-dataset comparisons

SD is the most common measure of variability/dispersion for a single data set, but it is not useful when comparing two different data sets. In such cases, CV is the method of choice

For instance, consider two different data sets:

- Data 1: Mean1 = 120000; SD1 = 2000
- Data 2: Mean2 = 900000; SD2 = 10000

It seems that **Data 2 is more spread out than Data 1**

However, let us now calculate CV for both data sets:

- CV1 = SD1/Mean1x100 = 1.6%
- CV2 = SD2/Mean2x100 = 1.1%

We can conclude **Data 1 is more spread out than Data 2**

Metric Selection

Sum Squared Error (SSE)

Cumulative squared difference between actual and predicted values

$$SSE = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

Sum of Squared Errors

y_i = True value for observation i
$\hat{y}_i$ = Predicted value for observation i
n = Number of test observations
SSE = Sum of Squared Errors

Error Visualization:
(Actual - Predicted)^2 → Sum across all observations

- Squares ensure all errors are positive
- Penalizes larger errors more heavily
- Foundation for MSE and RMSE calculations

Mean Squared Error (MSE)

Average squared difference - most common regression metric

$$MSE = \frac{1}{n} \times \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

Mean Squared Error

$$MSE = \frac{SSE}{n}$$

Alternative Form

Low MSE	High MSE
Better predictions	Poor predictions
Small deviations	Large deviations

Standard metric for regression model evaluation and optimization

Root Mean Squared Error (RMSE)

Square root of MSE - interpretable in original units

$$RMSE = \sqrt{\left[\frac{1}{n} \times \sum_{i=1}^{n} (y_i - \hat{y}_i)^2\right]}$$

Root Mean Squared Error

$$RMSE = \sqrt{MSE}$$

Simplified Form

Calculate MSE → **Apply** Square Root → **Get** RMSE

Key Advantage: Same units as target variable
Interpretation: "Average prediction error in original units"

RMSE shares same unit as target variable, making it more interpretable for stakeholders

Metric Selection

⚠

MSE/RMSE Limitations

Critical considerations when using squared error metrics

$\text{MSE} \propto (\text{Large Error})^2$

Outlier Sensitivity

One large error ≫ Many small errors

Disproportionate Impact

- **Sensitivity to Outliers:** Single large error dominates metric
- **Equal Weighting:** One big error = many small errors in impact
- **Scale Dependency:** Hard to compare across different problems
- **Non-Robust:** Not suitable for datasets with frequent outliers

Alternative: Mean Absolute Error (MAE) = $\frac{1}{n} \times \sum|y_i - \hat{y}_i|$
Advantage: MAE is more robust to outliers

Recommendation: Use MSE/RMSE for normal errors, MAE for outlier-prone data

- The **lower the MSE/RMSE** value, the **better the model** is with predictions
- A **high MSE/RMSE** indicates that there are **large deviations between the predicted and actual values**

Model - One				
Observations	Actual Value in $ (Ŷ)	Predicted Value in $ (Y)	Y - Ŷ	$(Y - \hat{Y})^2$
1	8	7	-1	1
2	6	7	1	1
3	4	5	1	1
4	3	2	-1	1
5	7	8	1	1
6	8	7	-1	1
7	9	8	-1	1
8	4	5	1	1
9	2	1	-1	1
10	1	2	1	1
		MSE	10 ÷ 10 = 1	

Model - Two				
Observations	Actual Value in $ (Ŷ)	Predicted Value in $ (Y)	Y - Ŷ	$(Y - \hat{Y})^2$
1	8	8	0	0
2	6	6	0	0
3	4	4	0	0
4	3	3	0	0
5	7	7	0	0
6	8	8	0	0
7	9	9	0	0
8	4	4	0	0
9	2	2	0	0
10	1	4.1622777	3.1622777	10
		MSE	10 ÷ 10 = 1	

The predictions of Model-One had an error of 1 for all 10 samples; for Model-Two, 9 samples were predicted correctly, but one prediction had an error of 4.16 implies **the single big error made by Model-Two has the same impact as the 10 small errors made by Model-One**

Metric Selection

Mean Absolute Error (MAE)

Average absolute difference between actual and predicted values

$$MAE = (1/n) \times \sum|y_i - \hat{y}_i|$$

Mean Absolute Error Formula

y_i = Actual value for observation i
$\hat{y}_i$ = Predicted value for observation i
n = Number of observations
|·| = Absolute value operator

MAE Advantage	MSE Disadvantage
Big errors don't overpower small errors	Single large error dominates metric

- Provides relatively unbiased performance assessment
- Same units as target variable (interpretable)
- Robust to outliers compared to MSE/RMSE
- Linear penalty for errors

Mean Absolute Percentage Error (MAPE)

Average absolute percentage difference between actual and predicted values

$$MAPE = (100\%/n) \times \sum|(y_i - \hat{y}_i)/y_i|$$

Mean Absolute Percentage Error Formula

Advantage: Scale-independent (percentage units)
Interpretation: "Average prediction error as percentage of actual values"
Limitation: Undefined when $y_i = 0$

- Expressed as percentage for easy interpretation
- Enables comparison across different scales and units
- Popular in business and forecasting applications
- Asymmetric penalty (overestimation vs underestimation)

Relative Absolute Error (RAE)

Ratio of model errors to simple benchmark errors

$$RAE = \sum|y_i - \hat{y}_i| / \sum|y_i - \bar{y}|$$

Relative Absolute Error Formula

RAE Range: 0 to ∞ (0 = ideal case, lower values = better)

$\bar{y}$ = Mean of actual values (simple benchmark)
Denominator: Errors of simple mean predictor
Interpretation: How much better than simple mean prediction

- **RAE < 1:** Model outperforms simple mean prediction
- **RAE = 1:** Model performs same as mean prediction
- **RAE > 1:** Model performs worse than mean prediction
- **Use Case:** Comparing models with different error units

Key Advantage: Unitless metric enabling cross-model comparison

Metric Selection

R^2 Definition & Purpose

Coefficient of Determination (R^2): Statistical measure that determines how well a regression model predicts an outcome

$$R^2 = 1 - SS_{res}/SS_{tot}$$

Coefficient of Determination Formula

SS_{res} = Sum of Squares of Residuals = $\Sigma(y_i - \hat{y}_i)^2$

SS_{tot} = Total Sum of Squares = $\Sigma(y_i - \bar{y})^2$

y_i = Actual values

$\hat{y}_i$ = Predicted values

$\bar{y}$ = Mean of actual values

R^2 represents the proportion of variance in the dependent variable that is predictable from the independent variables

R^2 Interpretation Scale

Understanding what different R^2 values mean for model performance

$R^2 = 0$	$0 < R^2 < 1$	$R^2 = 1$
No predictive power	Partial prediction	Perfect prediction

$R^2 = 0$ → Model explains none of the variability
$0 < R^2 < 1$ → Model explains partial variability
$R^2 = 1$ → Model explains all variability

- **$R^2 = 0$:** Model predictions are no better than using the mean
- **$0 < R^2 < 1$:** Model captures some but not all patterns
- **$R^2 = 1$:** Model perfectly fits all data points

Scientific Interpretation

Deep understanding of R^2's statistical meaning

$$R^2 = \text{Explained Variation / Total Va...}$$

Variance Explanation Ratio

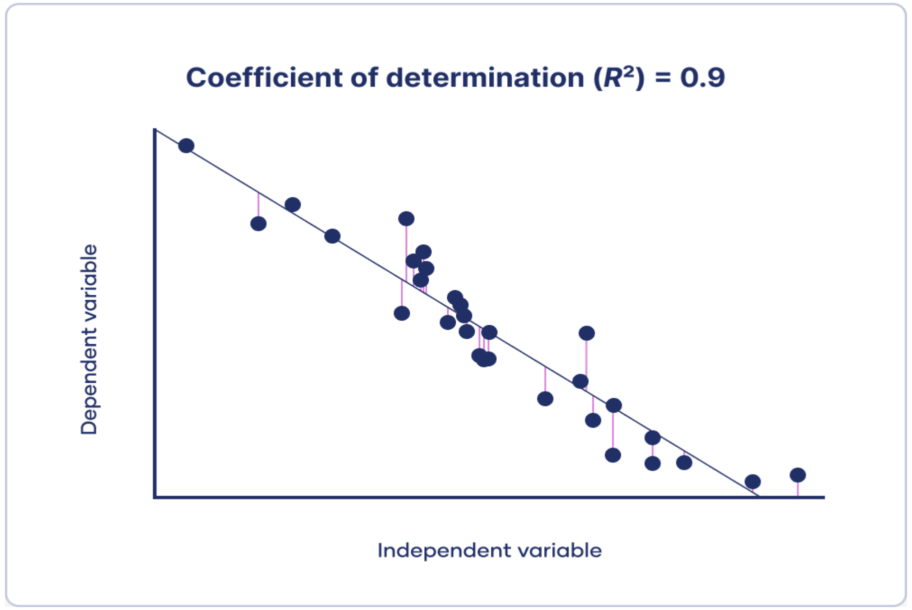
$$R^2 = \Sigma(\hat{y}_i - \bar{y})^2 / \Sigma(y_i - \bar{y})^2$$

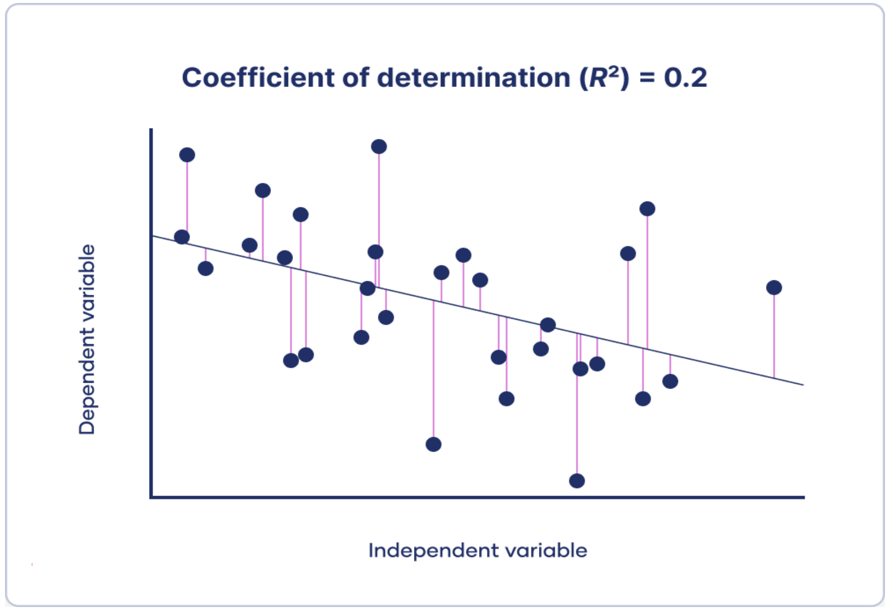
Alternative Calculation

- **Explained Variation:** How much of total variation is captured by model
- **Total Variation:** Natural variability in the data
- **Unexplained Variation:** Residual variation not captured by model

Key Insight: R^2 measures goodness-of-fit, not necessarily prediction accuracy
Caution: High R^2 doesn't guarantee good out-of-sample predictions
Best Practice: Use adjusted R^2 for multiple regression models

Metric Selection





R^2 is high, the observations are close to the model's predictions, that is, **most points are close to the line of best fit**

R^2 is low, the observations are far from the model's predictions, that is, **many points are far from the line of best fit**

Metric Selection

Metric	Advantages	Disadvantages
MAE	Easy to interpret and understand. Less sensitive to outliers.	Does not take into account the direction of errors.
MSE	Penalizes larger errors more heavily, giving it more sensitivity to outliers.	Harder to interpret than MAE, as it is not in the same unit as the original data.
RMSE	Has the same units as the original data, making it easier to interpret.	Sensitive to outliers.
R-squared	Easy to interpret, as it measures the proportion of the variance in the dependent variable that is explained by the independent variables.	Cannot distinguish between linear and nonlinear relationships. Can be misleading if used without considering the underlying assumptions.
Adjusted R-squared	Takes into account the number of independent variables in the model, making it more reliable than R-squared when comparing models with different numbers of variables.	Can be misleading if used without considering the underlying assumptions.
MAPE	Easy to interpret, as it gives the percentage error.	Not defined when the actual values are zero. Can be infinite when the actual value is very small.
WAPE	Takes into account the relative size of the errors, giving more weight to larger errors.	Can be influenced by outliers.
MASE	Takes into account the errors of the naive forecast, making it more appropriate for comparing models.	Not well known and less commonly used.

Model Validation

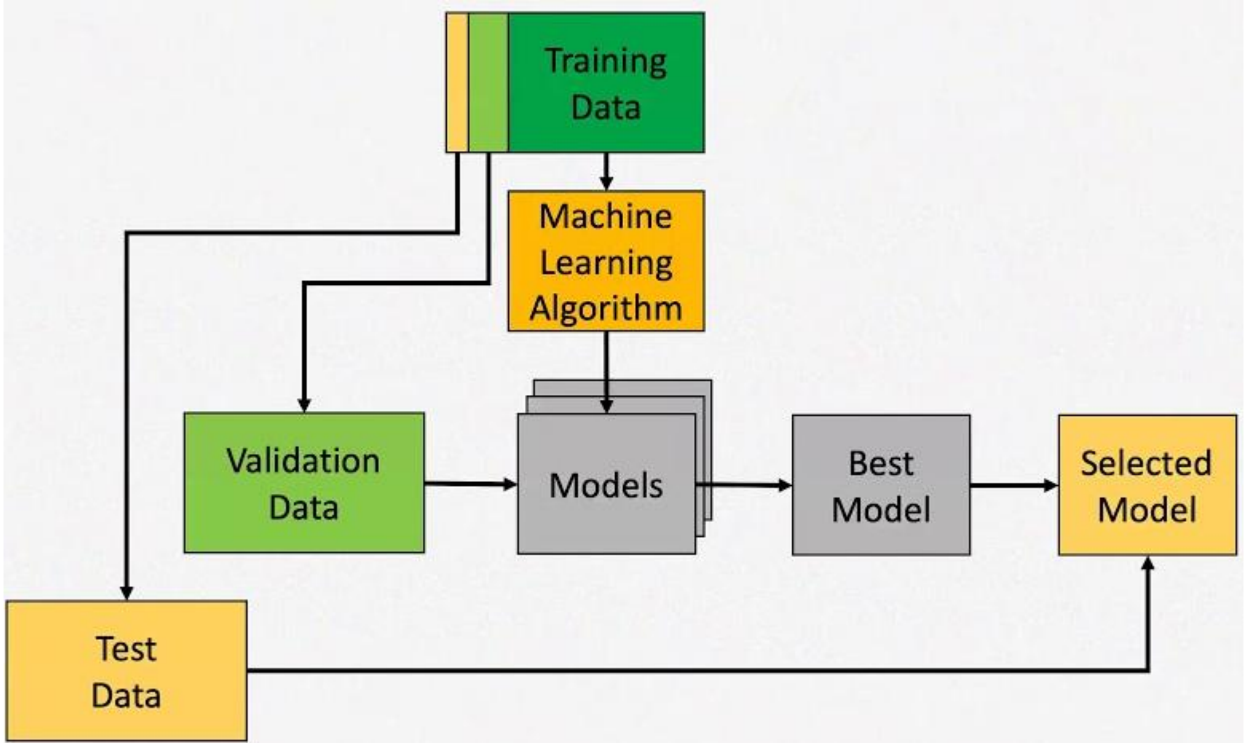
Model Validation

We seek answers to the following questions:

- how to assess the results of any classifier/regressor for the given problem?
- how to make sure that the results provided by the classifier/regressor are not artificial and will hold true when the model faces unseen data?

Given a labeled data set, we want to create random subsets (**training, validation, test**),

- as **independent** as possible (to minimize overlap between them)
- as **large** as possible (to obtain robust results)
- as **stratified** as possible (to maintain the original proportions of samples per class)



Source: www.cogitotech.com

Model Validation

Given a labeled data set, we have to divide it into three random subsets:

- **training set** (*learning*), to create the model
- **validation set** (learning), to optimize hyperparameters of the model (e.g., k of the k-NN, maximum depth of a decision tree, regularization factor of a SVM, etc.)
- **test set** (evaluation), to assess the model with data not used in the creation of the model

How many samples for learning and how many for testing?

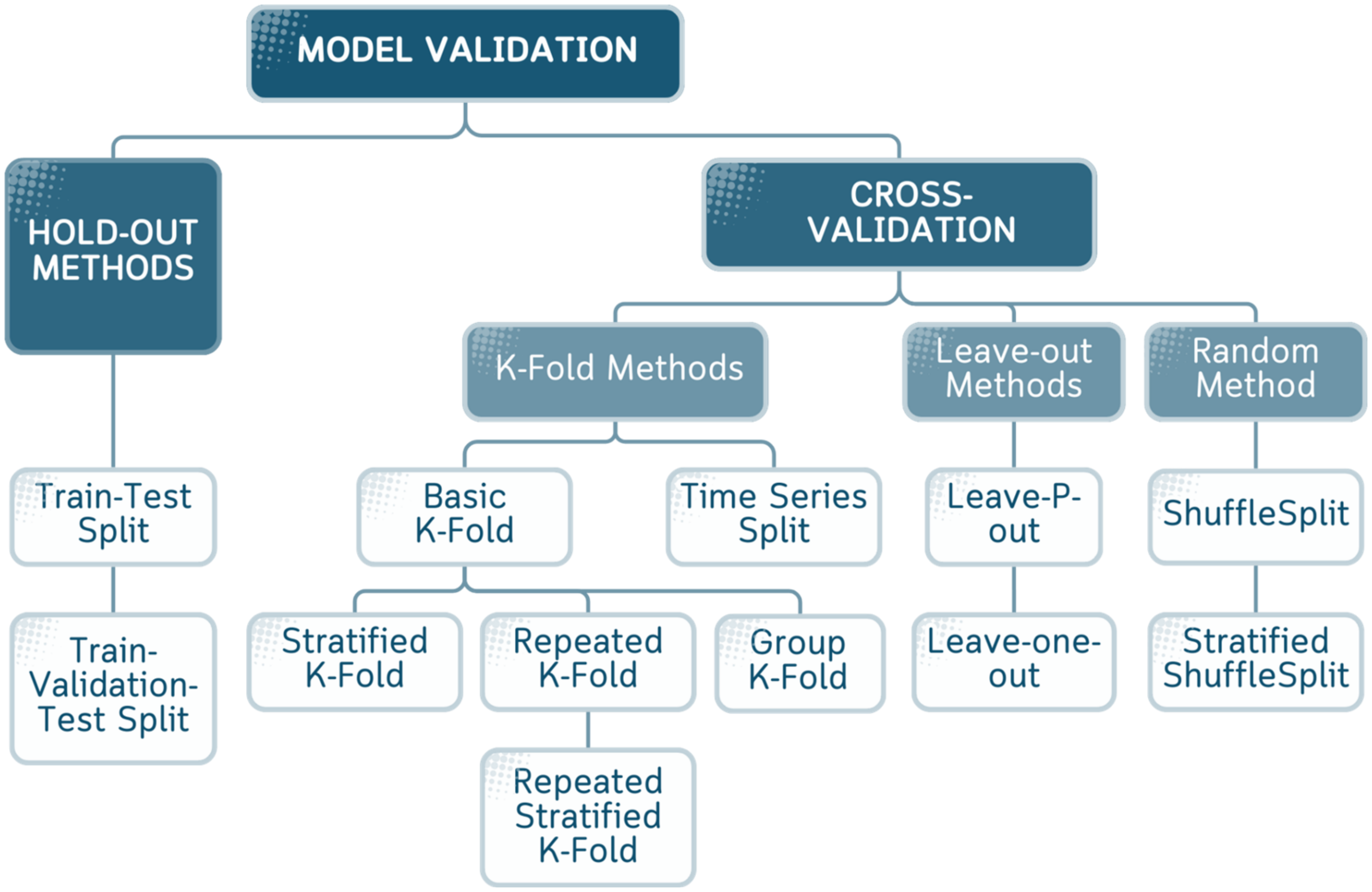
- if we have many samples to learn and few to evaluate, it can lead to **overfitting**
- if we have many samples to evaluate and few to learn, it can lead to **underfitting**

Thus, the idea is to **resample** the whole data set in a way that we have enough training samples to approximate the **generalization error**, tune the **hyperparameters** controlling the model complexity, and reduce the **test error**

Model Validation

Resampling methods for validating models:

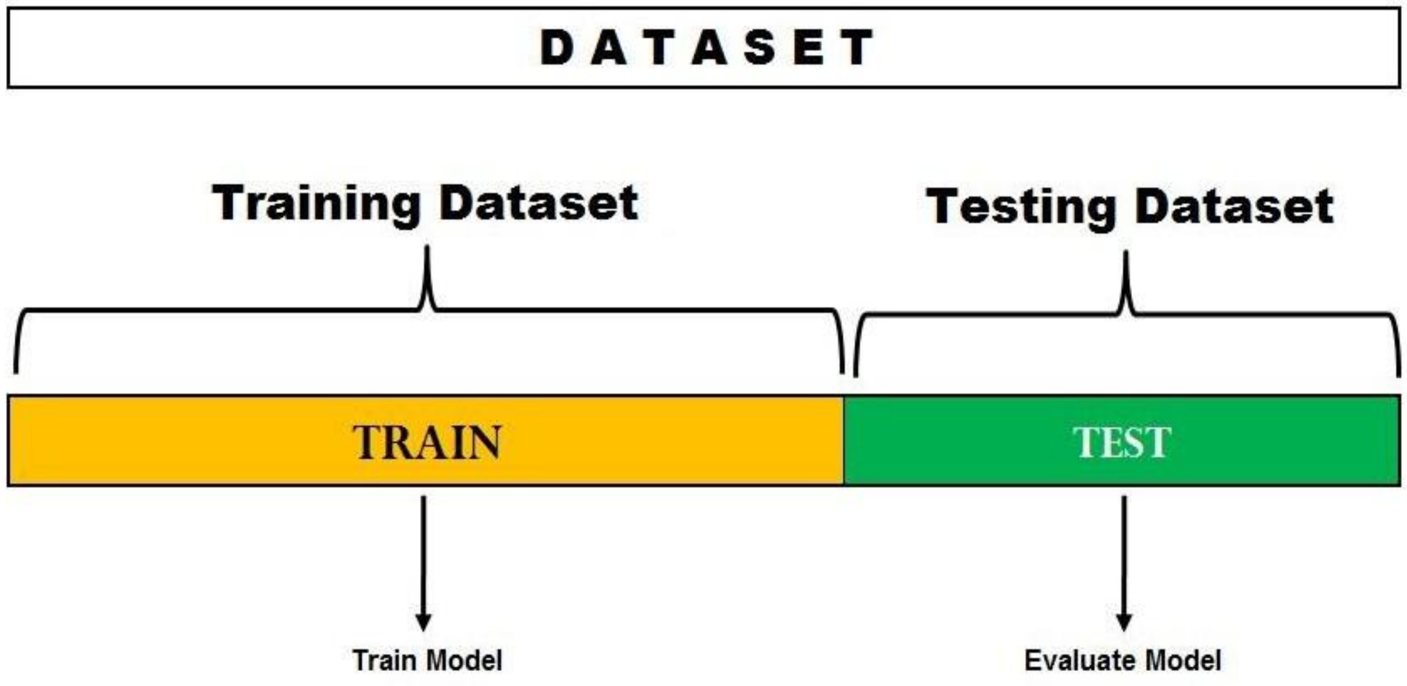
- arbitrarily large sets (rare case)
  - holdout
- moderate or small size set (common case)
  - K-fold cross-validation
  - leaving-one-out
  - bootstrapping



Source: www. towardsdatascience.com

Holdout

- Common ratios used for splitting *X* are **60:40, 70:30, 80:20**
- We can shuffle the data K different times and repeat the
- Process for each shuffled data set (repeated holdout)

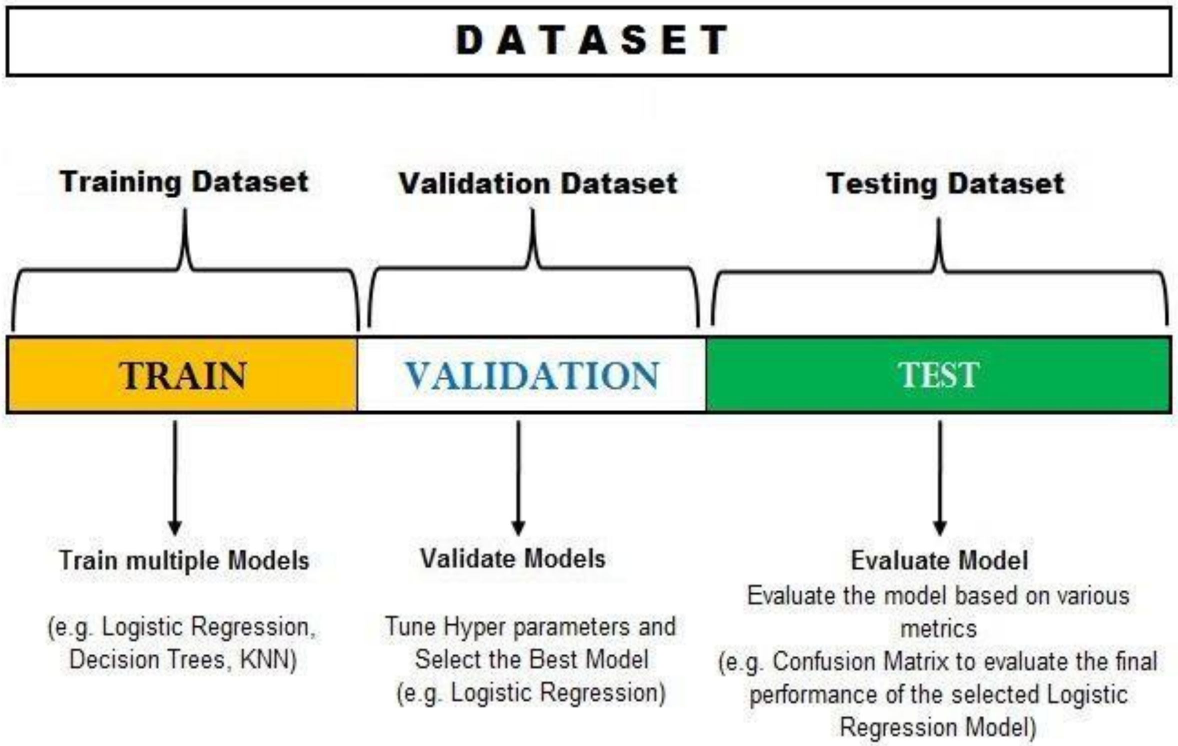


A Variation of Holdout

Holdout is not effective for comparing multiple models and tuning their hyperparameters

Solution: splitting of data into not two, but **three disjoint sets:** training, validation and test

- the training block is divided into **a portion for training and another for validation**
- the **validation set** is used **to tune the hyperparameters** and **select the best performing algorithm**
- the **test set** is used to **evaluate** the final selected model

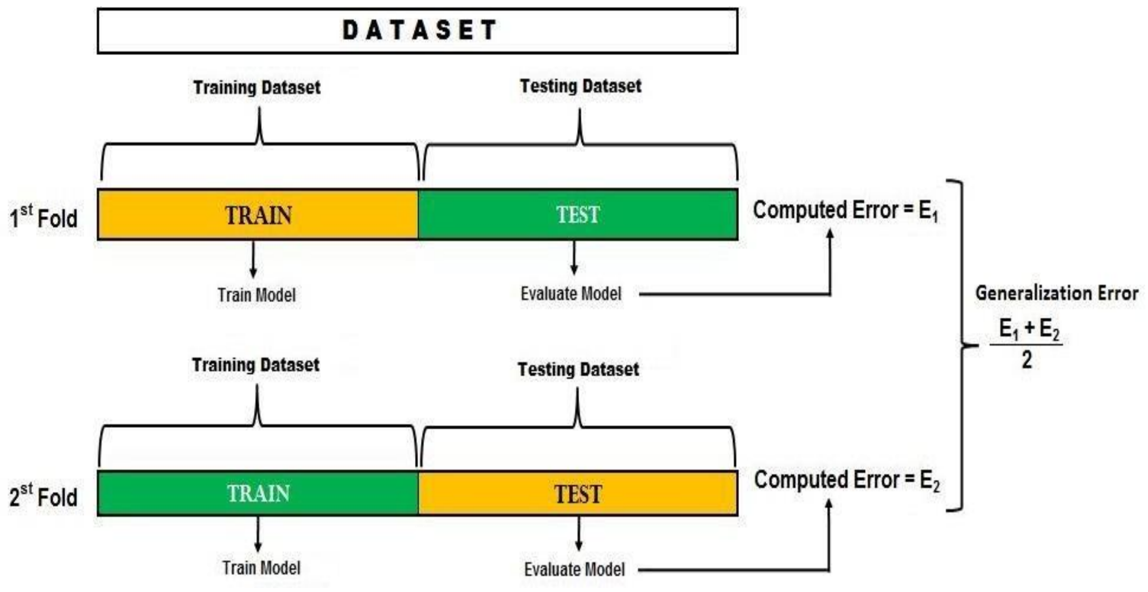


K-fold cross-validation

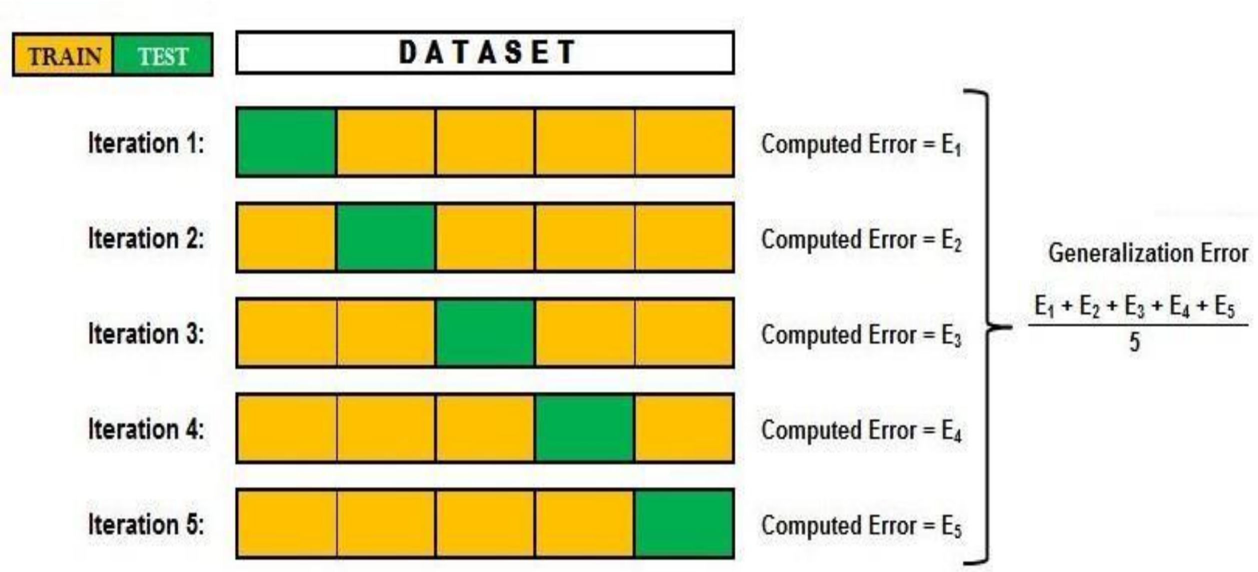
- Given a data set X, we randomly divide X into K equal sized blocks and generate K training sets and K test sets
- We leave out a part and train the model on the other K–1 blocks, using the left part to evaluate the model
- This process is repeated K times so that each part is used as testing set
- The results from each fold are then combined and
- Averaged to come up with the final error
- Typical values of K are 2, 5 and 10

K-fold cross-validation

Example (2-fold cross-validation)



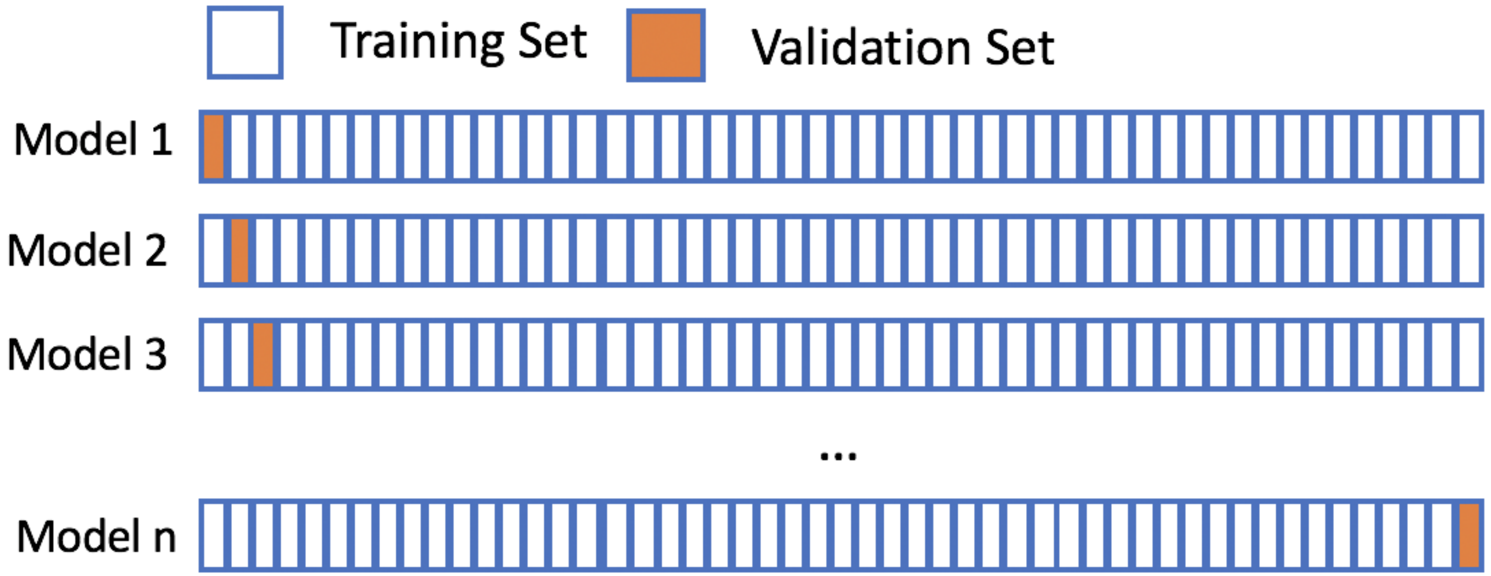
General case (e.g. 5-fold cross-validation)



Leaving-one-out

Given a data set *X* with *n* samples, the **leaving-one-out method is a particular case of K-fold cross-validation:**

- We divide the whole data set X into **n blocks**
- At each iteration, **one** sample is **for testing** and the remaining **n–1** samples are used **to train the model**
- We then **average the n results** and consider this value as our generalization error



THANK YOU

Any Questions?

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